

Phase I/II Operating Characteristics

Cycle 2 shared additive versus cycle 2 and cycle 3 multicycle additive

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.3	0.4	0.5	0.5	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
OBD selection (%)						
C2 shared additive	8.7	6.2	2.8	0.7	0.6	81
C2 multicycle additive	5.7	0.9	0.3	0	0	93.1
C3 multicycle additive	4.5	0.3	0.1	0.1	0	95
Mean administrations by dose						
C2 shared additive	10.9	4.4	2.5	1	0.3	
C2 multicycle additive	10.8	2.7	1	0.4	0.1	
C3 multicycle additive	10.5	2.6	0.8	0.2	0.1	
Scenario 16						
DLT rate	0	0.1	0.1	0.1	0.3	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
C2 shared additive	25.9	21.1	17.5	20	14.6	0.9
C2 multicycle additive	35.4	18.8	24.3	13.3	5.7	2.5
C3 multicycle additive	39.1	25.6	12.5	15.2	3.9	3.7
Mean administrations by dose						
C2 shared additive	7.1	6.6	5.7	5.8	4.9	
C2 multicycle additive	8.6	5.7	6.6	5.4	3.5	
C3 multicycle additive	8.9	7	5.2	5.7	3	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 20						
DLT rate	0.1	0.1	0.3	0.4	0.5	
Efficacy rate	0.1	0.2	0.2	0.2	0.2	
OBD selection (%)						
C2 shared additive	5.8	46.7	29.7	11.2	2.1	4.5
C2 multicycle additive	9.9	45.5	16.5	2.1	0.4	25.6
C3 multicycle additive	12.2	42.2	7.3	1.5	0	36.8
Mean administrations by dose						
C2 shared additive	5	10.3	7.9	4.7	1.6	
C2 multicycle additive	7.5	10.4	6.4	2.6	0.7	
C3 multicycle additive	8.2	10.9	4.8	2.2	0.6	
Scenario 24						
DLT rate	0	0	0.1	0.1	0.2	
Efficacy rate	0.1	0.1	0.2	0.5	0.5	
OBD selection (%)						
C2 shared additive	1.2	1.4	5.7	56.6	35.1	0
C2 multicycle additive	1.2	1.9	17.3	56.6	22.4	0.6
C3 multicycle additive	2.4	3.5	7.6	73.6	11.4	1.5
Mean administrations by dose						
C2 shared additive	3.4	3.4	4.2	10.7	8.2	
C2 multicycle additive	3.8	3.6	5.9	10.3	6.4	
C3 multicycle additive	4	4.5	4.6	12.1	4.7	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.1	0.2	0.3	0.5	0.6	
Efficacy rate	0.1	0.2	0.4	0.4	0.5	
OBD selection (%)						
C2 shared additive	10.5	22.3	47.2	10.9	1.2	7.9
C2 multicycle additive	23.5	23.1	22.2	1.8	0.1	29.3
C3 multicycle additive	25.9	25.1	9.8	1.2	0	38
Mean administrations by dose						
C2 shared additive	6.8	7.5	9.7	4.1	1.1	
C2 multicycle additive	10.3	7.7	6.5	2.2	0.4	
C3 multicycle additive	11.4	8.1	4.7	1.7	0.3	
Scenario 38						
DLT rate	0.1	0.1	0.1	0.1	0.3	
Efficacy rate	0.1	0.1	0.1	0.1	0.5	
OBD selection (%)						
C2 shared additive	4.6	6.8	7.7	12	64.6	4.3
C2 multicycle additive	7.8	12.2	18.5	15.5	29.1	16.9
C3 multicycle additive	11	13.7	15	17.2	18.5	24.6
Mean administrations by dose						
C2 shared additive	4.4	4.4	4.9	5.5	10.7	
C2 multicycle additive	5.9	5.5	6	5.3	6	
C3 multicycle additive	6.6	6.5	5.3	5.4	4.5	

Mean Unique Patients by Method

Mean number of unique patients per simulated trial

Scenario	1	16	20	24	27	38	Mean
C2 shared additive	12.4	18.3	19	19.7	19.6	18.9	18
C2 multicycle additive	10	19.6	18	20.6	18	18.6	17.5
C3 multicycle additive	8.3	16.8	14.8	18.1	14.9	15.1	14.7

Values are reported from Total_Unique_Patients and shown to one decimal. The Mean column is the unweighted mean across the six scenarios.

All rows use N = 30, a 0.30 target toxicity, a 0.20 efficacy threshold, and a 0.85 futility cutoff.